

Project: Decision Tree Classification Model

Part 1: SalesBoost CRM Analytics

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1. Business Overview

SalesBoost CRM Analytics is a business-to-business software company that sells a cloud-based CRM and sales automation platform to small, medium, and large organizations across Canada. The company receives leads from several channels including LinkedIn Ads, Google Search, webinars, referrals, trade shows, email campaigns, and organic website traffic.

The sales team must decide how to prioritize follow-up because sales representatives have limited time. Some leads are highly likely to become paying customers, while others require more education, a different offer, or lower-priority follow-up. A Decision Tree classification model helps the business identify patterns in lead behavior and sales readiness, enabling smarter prioritization decisions.

Element	Description
Business	SalesBoost CRM Analytics
Objective	Predict which B2B leads will become won deals
Target Variable	Deal_Won (Yes / No)
Dataset Size	264 records (260 after removing 4 duplicates)
Features	20 input features (9 numerical, 11 categorical)
Priority Metric	F1-Score — balances missed revenue (FN) and wasted sales time (FP)
Model Used	Decision Tree Classifier (sklearn, max_depth=4, random_state=42)

2. Data Preparation

2.1. Dataset Overview

The dataset contains 264 records representing B2B leads in the SalesBoost sales pipeline. After removing 4 duplicate rows, 260 clean records were used. Each record has 23 original columns; 20 were retained as input features after dropping Lead_ID, Lead_Date, and the target variable Deal_Won. Features fall into four business categories:

Feature Group	Columns
Lead & Company Profile	Lead_Source, Industry, Region, Company_Size, Employee_Count, Annual_Revenue_Estimate, Use_Case
Engagement Behavior	Website_Visits_30D, Email_Open_Rate, Demo_Requested
Sales Process Activity	Sales_Calls_Count, Proposal_Sent, Budget_Confirmed, Decision_Maker_Involved, Competitor_Considered
Commercial & Pipeline	Discount_Offered_Percent, Days_In_Pipeline, Estimated_Deal_Value, Lead_Score, Sales_Rep

2.2. Class Distribution

The target variable Deal_Won was well-balanced across the full dataset, supporting reliable model training without class imbalance adjustments:

Class	Full Dataset (260)	Train / Test
Yes (Deal Won)	136 (52.3%)	109 / 27
No (Deal Not Won)	124 (47.7%)	99 / 25

2.3. Preprocessing Steps

- Dropped Lead_ID (unique identifier with no predictive value) and Lead_Date (date string; no time-series modelling applied).
- Removed 4 duplicate rows. The dataset reduced from 264 to 260 records.
- Detected missing values across 7 columns (e.g. Annual_Revenue_Estimate: 7, Email_Open_Rate: 6, Industry: 4, Website_Visits_30D: 4). Numerical missing values filled with column median; categorical missing values filled with column mode.
- Confirmed zero missing values after imputation.
- Converted currency columns (Annual_Revenue_Estimate, Estimated_Deal_Value) and percentage column (Email_Open_Rate) from string format to numeric float by removing \$ and % characters.
- Applied One-Hot Encoding (sklearn ColumnTransformer + OneHotEncoder) to all 11 categorical features, expanding the feature space from 20 to 54 processed features.
- Numerical features (9 columns) passed through unchanged, no scaling applied, as Decision Trees do not require feature scaling.
- Split data 80/20 using train_test_split with stratify=y and random_state=42, producing 208 training records and 52 testing records.

3. Decision Tree Model

3.1. How Decision Trees Make Predictions

A Decision Tree is a supervised machine learning algorithm that recursively partitions data by asking binary questions about feature values. At each node, the algorithm selects the feature and threshold that best separates lead that became won deals from those that did not, using Gini impurity as the splitting criterion.

For SalesBoost CRM Analytics, the tree learns business rules such as: 'If Lead_Score > 58 AND Estimated_Deal_Value > \$20,000, predict Deal_Won = Yes.' These rules are directly interpretable; a sales manager can follow any prediction path through the tree and understand exactly which factors the model responded to.

3.2. Decision Tree Visualization

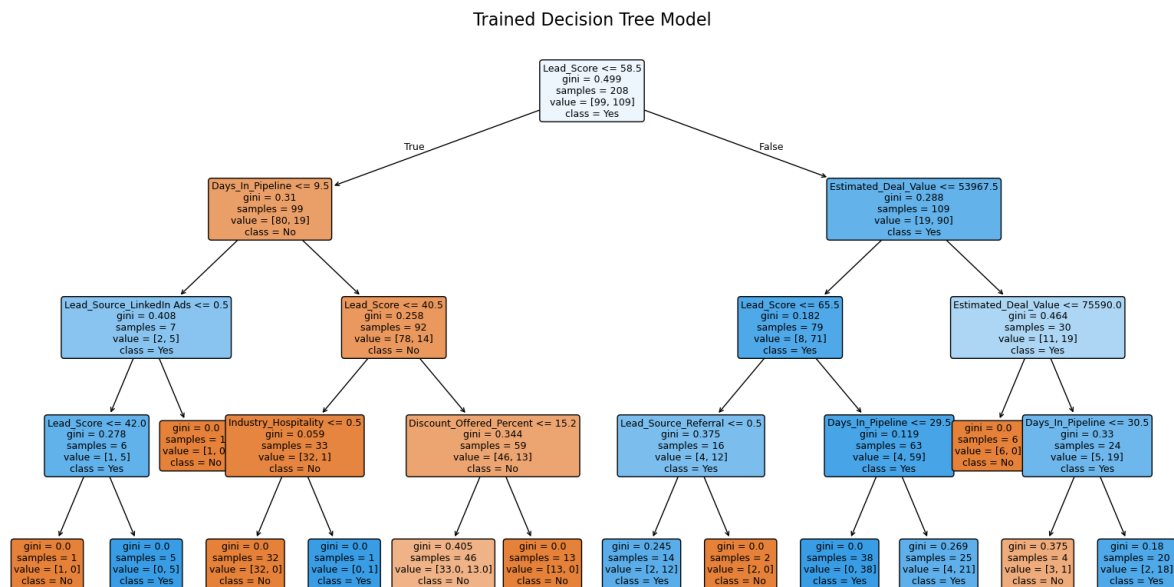


Figure 1: Decision Tree Model, SalesBoost CRM Analytics

The decision tree was trained using a maximum depth of 4, meaning it makes at most four sequential splitting decisions before reaching a final prediction. The tree is displayed with colour-filled nodes, where the shading indicates the dominant class at each node (darker shading toward one colour reflects greater class purity (a higher proportion of either "Yes" or "No" outcomes)).

Starting from the root node at the top, the tree selects the single feature that best separates won deals from lost ones and uses it as the first split. At each subsequent level, the

tree continues subdividing the data using the most informative remaining features. Each internal node displays the feature name, the threshold or category used for the split, the Gini impurity (a measure of how mixed the classes are 0 means perfectly pure), the number of samples reaching that node, and the class distribution at that point.

The leaf nodes at the bottom represent the model's final predictions. A leaf with low Gini impurity and a strong majority of one class indicates a confident prediction. Leaves predicting "Yes" (deal won) tend to be associated with combinations of high-engagement signals, such as a demo being requested, budget being confirmed, or decision-maker involvement while "No" leaves reflect leads lacking these indicators.

3.3. *Model Configuration*

Parameter	Value / Rationale
max_depth	4: limits tree depth to prevent overfitting while retaining predictive power
random_state	42: ensures reproducible results across runs
criterion	Gini (default): measures node impurity for splitting decisions
Tree depth	4 levels
Leaf nodes	14 (original tree) / 12 (controlled tree with min_samples_leaf=5)
Processed features	54 (after one-hot encoding of 11 categorical features)

4. **Model Evaluation**

4.1. *Confusion Matrix*

The confusion matrix below shows the breakdown of predictions on the test set (52 records). The rows represent the actual class, what the lead actually did. The columns represent the predicted class, what the model said they would do.

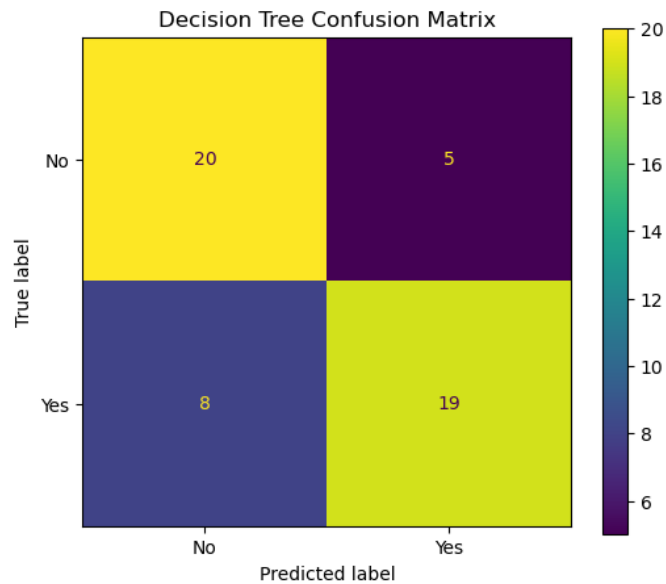


Figure 2: Confusion Matrix, SalesBoost CRM Analytics (max_depth=4 model)

Reading the matrix:

- **True Negatives (20):** The model correctly predicted 20 leads would not become won deals. The sales team was right to avoid prioritizing these leads, saving time and resources.
- **False Positives (5):** The model predicted 5 leads would become won deals, but they did not. Sales reps may have spent time on proposals and follow-up calls for leads that were not ready to buy.
- **False Negatives (8):** The model predicted 8 leads would not convert, but they actually would have. These are missed revenue opportunities, leads that may have been under-prioritized or lost to a competitor.
- **True Positives (19):** The model correctly identified 19 leads who became won deals. These were successfully prioritized, resulting in direct revenue for SalesBoost.

A false positive means the model predicted a lead would become a won deal, but it did not. There were 5 false positives in the test set. In practice, sales reps would be directed to prioritise these leads, scheduling follow-up calls, preparing tailored proposals, and potentially offering discounts, for prospects who were never going to convert. The direct cost is wasted time and

resources. The indirect cost is opportunity cost: hours spent on these non-converting leads are hours not spent on genuinely promising ones.

A false negative means the model predicted a lead would not become a won deal, but it actually would have. There were 8 false negatives, the more common and more damaging error type. These are missed opportunities: a buyer with real purchase intent gets placed in a low-priority nurturing queue, receives slow or minimal follow-up, and may sign with a competitor before SalesBoost's sales team circles back. Given that the business plan identifies this exact scenario; "high-potential leads may not receive fast follow-up", false negatives represent the most strategically costly failure mode of the model.

4.2. *Classification Metrics*

The model achieved **75% accuracy** on 52 unseen test leads. Breaking that down through the confusion matrix: it correctly identified 19 out of 27 actual wins (true positives) and correctly flagged 20 out of 25 actual losses (true negatives). Precision for the positive class was 79.2%, recall was 70.4%, and the F1-score was 74.5%. For a synthetic dataset of 260 records, this is a reasonable and meaningful baseline considerably better than random guessing (which would be around 50%), though there is clear room to improve, particularly on recall.

Metric	Result	Business Meaning
Training Accuracy	89.4%	High, expected on data the model was trained on
Testing Accuracy	75.0%	3 in 4 unseen leads correctly classified
Train / Test Gap	~14.4 pts	Mild overfitting, model generalises reasonably
Precision (Yes)	79.2%	Of leads flagged as wins, 79% won
Recall (Yes)	70.4%	Model identified 70% of all actual wins
F1-Score (Yes)	74.5%	Balanced score, the priority metric
False Positives	5	Leads incorrectly flagged as wins, wasted sales effort
False Negatives	8	Real wins missed, potential revenue lost

4.3. *The Important Evaluation Metric*

Recall is the most critical metric for SalesBoost. The business plan explicitly warns that "high-potential leads may not receive fast follow-up" and that slow response risks losing customers to competitors. Recall measures what proportion of actual won deals the model

correctly flagged, a low recall means real opportunities are being silently classified as low-priority and left to go cold. That said, neither metric can be ignored in isolation: F1-score is the best single metric to optimise when both wasting rep time (precision) and missing opportunities (recall) carry business costs, which is the case here. With a recall of 70.4%, the model currently misses about 3 in every 10 genuine winners, a meaningful gap for a sales team.

4.4. *Overfitting Analysis*

The model shows mild-to-moderate signs of overfitting. The primary evidence is the 14-point train/test accuracy gap. As `max_depth` increases from 1 to 10, training accuracy climbs steadily from 81.7% to 100%, while testing accuracy peaks around depth 1–2 (82.7%) and then fluctuates or declines. This is the classic overfitting signature; the model gets better and better at memorising training examples while failing to generalise. The unrestricted tree (`max_depth=None`) demonstrates the extreme case: 100% training accuracy collapses to only 69.2% on the test set, a 30-point drop. The controlled tree (`max_depth=4`, `min_samples_leaf=5`) partially addresses this, training accuracy comes down to 86.5% while testing accuracy is preserved at 75.0%, a healthier trade-off.

Max Depth	Training Accuracy	Testing Accuracy	Train/Test Gap
1	81.7%	82.7%	–1.0 pts
2	83.2%	82.7%	0.5 pts
3	86.5%	75.0%	11.5 pts
4	89.4%	75.0%	14.4 pts
5	92.3%	75.0%	17.3 pts
7	95.7%	71.2%	24.5 pts
10	100.0%	69.2%	30.8 pts

5. **Controlling Decision Tree Complexity**

5.1. *Training Accuracy vs Testing Accuracy*

There is a large difference between training accuracy and testing accuracy. The original Decision Tree (`max_depth=4`) achieved 89.4% on training versus 75.0% on testing, a difference of approximately 14 percentage points. This tells us the model has learned the training data somewhat too specifically. Adding `min_samples_leaf=5` further reduced leaf count from 14 to

12 and the train/test gap from 14.4 to 11.5 points, representing the best balance of simplicity and performance across all configurations tested.

Comparison	Unconstrained	Original (depth=4)	Controlled (depth=4, leaf=5)
Training Accuracy	100.0%	89.4%	86.5%
Testing Accuracy	69.2%	75.0%	75.0%
Train/Test Gap	30.8 pts	14.4 pts	11.5 pts
Tree Depth	10	4	4
Leaf Nodes	35	14	12

6. Feature Importance

Feature importance measures how much each feature contributed to reducing Gini impurity across all splits in the Decision Tree. Features with higher scores were used more frequently and at higher levels of the tree, making them the primary drivers of the model's predictions.

The remaining 47 out of 54 features contributed zero importance to the model. This is a significant finding, the tree's decisions are almost entirely driven by just three continuous variables: the composite lead score, the estimated deal size, and how long a lead has been sitting in the pipeline.

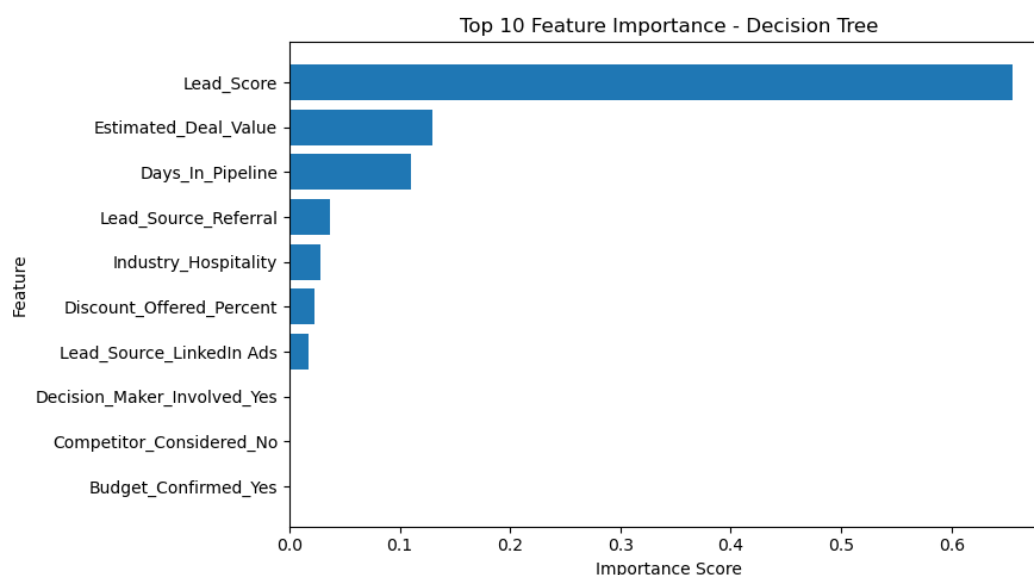


Figure 3: Top Important Features, SalesBoost CRM Analytics

6.1. Business Value of the Most Important Features

Lead_Score (65.5%): This single feature dominates the model. SalesBoost should ensure their lead scoring methodology is rigorous, consistently applied, and regularly recalibrated. If the score is reliable, sales reps can use it as a first filter before doing anything else. Conversely, if the score is computed from incomplete CRM data or rep-subjective inputs, its dominance in the model is a risk, garbage in, garbage out at scale.

Estimated_Deal_Value (13.0%): Higher-value deals are more likely to be won. This gives managers a clear instruction: when two leads have similar scores, prioritize the one with the larger estimated deal value. It also signals that deals worth investing extra time and senior rep attention in are identifiable early.

Days_In_Pipeline (11.0%): Leads that have been in the pipeline too long are a warning sign. The business should consider setting follow-up time limits or escalation triggers, if a lead has exceeded a threshold number of days without progressing, it should either receive urgent attention or be moved to a different nurturing track.

Lead_Source_Referral (3.7%): Referral leads convert at a higher rate than other sources. Marketing budgets and partner programs that drive referrals are worth protecting or expanding.

Rank	Feature	Importance Score
1	Lead Score	65.5%
2	Estimated Deal Value	13.0%
3	Days In Pipeline	11.0%
4	Lead Source Referral	3.7%
5	Industry Hospitality	2.8%
6	Discount Offered Percent	2.3%
7	Lead Source LinkedIn Ads	1.7%

7. Business Actions and Recommendations

Model Insight	Recommended Business Action
Lead_Score is the dominant predictor (65.5%)	Ensure the lead scoring methodology is rigorous, consistently applied by all reps, and regularly recalibrated. Any inaccuracy in Lead_Score propagates directly into model quality. Consider auditing how scores are assigned, especially for edge cases.

Estimated_Deal_Value strongly predicts wins (13.0%)	When two leads have comparable scores, prioritize the one with the higher estimated deal value. Invest senior rep time proportionally to potential revenue. Set a deal-value threshold for automatic fast-track follow-up.
Days_In_Pipeline is a significant signal (11.0%)	Introduce pipeline age triggers: if a lead exceeds 30 days without a stage progression, escalate it for review or move it to a nurturing workflow. Time-stalled deals rarely close without renewed intervention.
Referral leads convert at above-average rates	Protect and grow the referral channel. Consider a formal referral incentive program for existing clients and partners. Referral leads should receive prompt, high-touch follow-up given their higher conversion likelihood.
F1-Score is 74.5%	The model has solid predictive value but is not yet strong enough for fully automated prioritization. Use it as a ranking tool: sort the pipeline by predicted probability and focus the first 70% of rep capacity on the top-ranked leads, with human review of borderline cases.
8 false negatives in 52 test records	Consider lowering the classification threshold (using predict_proba) to increase Recall at the cost of some Precision. If missing a genuine win is more costly than spending time on one weak lead, a threshold below 0.5 may improve business outcomes.

8. Limitations

8.1. Dataset

- **Synthetic dataset:** The dataset was synthetically generated for teaching purposes. Real CRM data contains human entry errors, inconsistent field definitions, and more complex, less evenly distributed patterns.
- **Small sample size:** 260 records is very small for a production-grade machine learning model. Each misclassification in the 52-record test set shifts accuracy by approximately 1.9 percentage points, making results sensitive to the random split. A minimum of several thousand records per campaign cycle would be needed for reliable deployment.

8.2. Model

- **Lead_Score dominance:** With 65.5% importance, the model is heavily dependent on a single derived field. If Lead_Score itself is computed from biased, incomplete, or subjective CRM inputs, the model will learn and amplify those upstream biases.

- **Lead_Date excluded:** Time-based patterns (seasonal trends, quarter-end effects, day-of-week behavior) were not captured. Incorporating temporal features derived from Lead_Date could meaningfully improve model performance.
- **Zero importance for most features:** 47 of 54 processed features contributed zero importance. Important qualifying signals: Budget_Confirmed, Decision_Maker_Involved, Demo_Requested, were unused by the tree, likely because Lead_Score already captured much of the same information.

9. Responsible AI Considerations

- **Human review before deployment:** Sales managers should review the model's top-ranked lead segments before acting on predictions. The model is a decision-support tool, it should inform, not replace, human judgment about relationship quality, strategic value, and unusual circumstances.
- **Fairness and bias audit:** Before any real-world use, the model should be tested for disparate performance across industry, region, and company size. If certain groups historically received less sales attention, the model may have learned to deprioritize them further.
- **Explainability:** The Decision Tree structure supports full explainability. The decision path for any individual lead prediction can be extracted and explained to managers and clients in plain language, a meaningful advantage over black-box models.
- **Model drift:** Lead conversion patterns change over time as market conditions, product offerings, and competitor landscapes evolve. The model should be retrained periodically on fresh CRM data to maintain accuracy and relevance.
- **Data quality dependency:** The model is only as good as the CRM data it is trained on. Incomplete records, inconsistently scored leads, or reps who skip CRM updates will

degrade model quality. Data quality governance is a prerequisite for reliable predictions.

Finally, human judgement should still be used when making business decisions as the model only knows what is in the CRM. It cannot see that a prospect's CFO just sent a warm follow-up email, that a competitor recently lost a major client and that company is now actively shopping, that a rep has a personal relationship with a decision-maker, or that a deal is strategically important regardless of its predicted probability. Beyond these blind spots, the model also cannot account for unusual circumstances, ethical considerations, or long-term relationship value that doesn't show up in structured fields. Sales managers should treat the score as one strong input alongside their own knowledge, flag cases where the model's prediction feels wrong, and periodically audit the model's recommendations to check that it is still fair, accurate, and aligned with how the business has evolved.